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ENBC 332

Homework 5 Discussion and Outputs

Problem 1 Part C:

For part A, since the sample size was very large, it was safe to assume that the confidence interval given by the z scores were accurate. For part B, this trend of approximating the confidence interval based on sample size is shown from least to greatest. More specifically, the greatest sample size being the most similar to the z score calculated confidence interval compared to the lowest sample size.

Within the context of the dataset, which is related to coffee sales by age, the confidence interval offers insight to the range of ages that are most frequent to purchase something from the coffee store. It tells us that the average age for a person to purchase coffee is between the ages of 38 and 41.

Based on the plot, it is obvious to notice that as the sample size increases, all of the confidence intervals converge and decrease the overall range. This rate of change demonstrates the effect that sample size has on the range of the confidence interval.

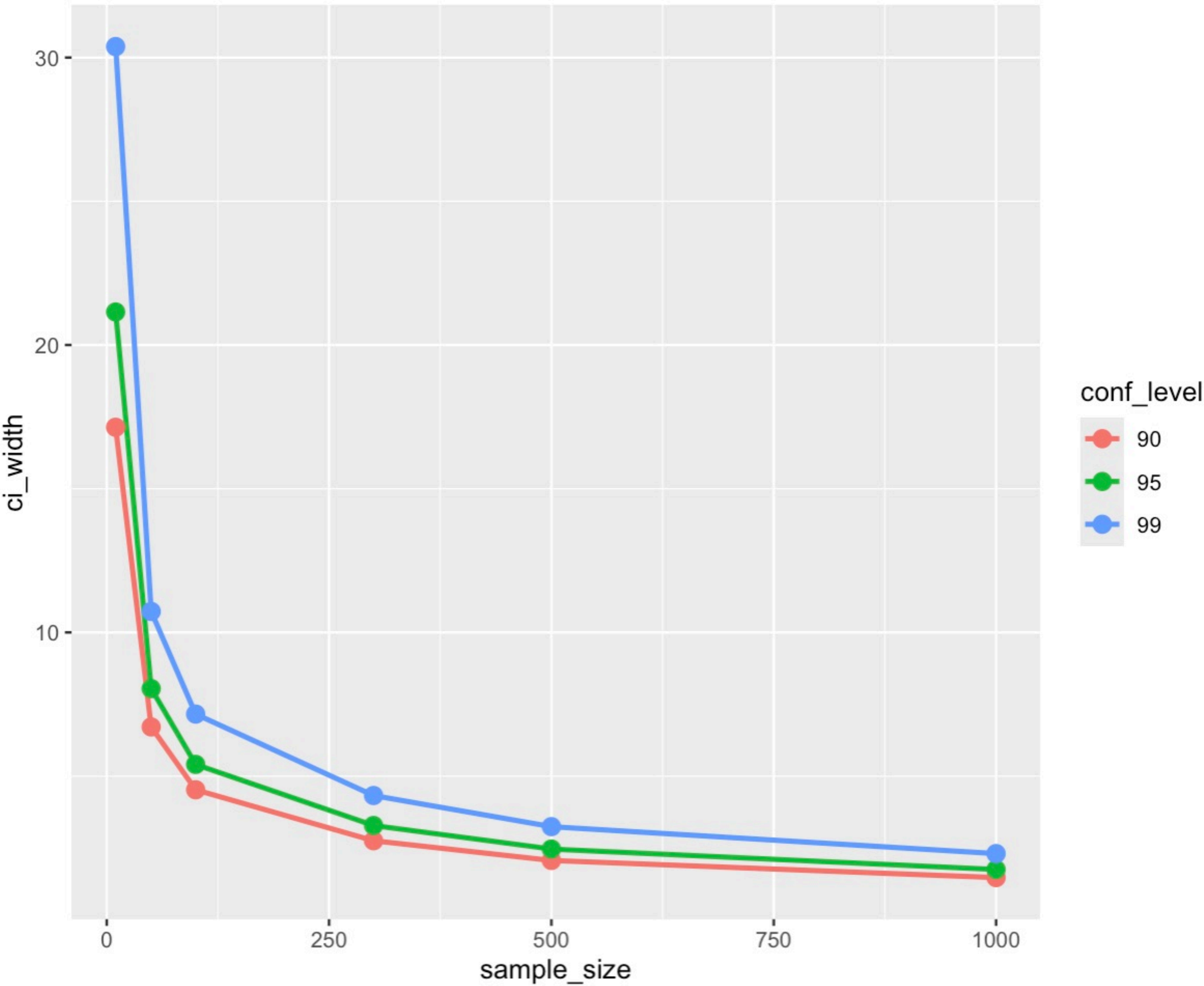
Outputs:

Problem 1:

```
> library(readxl)
> library(ggplot2)

 Learn more about the underlying theory at https://ggplot2-book.org/

> data <- read_excel('/Users/chrislee/Desktop/R Workspace/ENBC 332 - Homework2/ENBC 332 - HW02 Problem1 Excel File.xlsx', sheet = "Sheet1")
> data_age = data$age
> random_sample_values = sample(data_age, size=500, replace=FALSE)
> CI_func_zscore <- function(sample_data, conf_level) {
+   sample_mean <- mean(sample_data)
+   sample_size <- length(sample_data)
+   sample_sd <- sd(sample_data)
+ 
+   sample_standard_error <- sample_sd/sqrt(sample_size)
+   alpha <- 1 - conf_level
+   critical_z <- qnorm(1 - (alpha / 2))
+   sample_moe <- critical_z * sample_standard_error
+ 
+   lower_bound <- sample_mean - sample_moe
+   upper_bound <- sample_mean + sample_moe
+   return(cat("Confidence Interval for", conf_level * 100, "% \n[",lower_bound, upper_bound,"]\n"))
+ }
> CI_zscore_ninety <- CI_func_zscore(random_sample_values, 0.9)
Confidence Interval for 90 %
[ 37.77397 39.81003 ]
> CI_zscore_ninety_five <- CI_func_zscore(random_sample_values, 0.95)
Confidence Interval for 95 %
[ 37.57894 40.00506 ]
> CI_zscore_ninety_nine <- CI_func_zscore(random_sample_values, 0.99)
Confidence Interval for 99 %
[ 37.19776 40.38624 ]
> random_sample_ten <- sample(data_age, size=10, replace=FALSE)
> random_sample_fifty <- sample(data_age, size=50, replace=FALSE)
> random_sample_onehundred <- sample(data_age, size=100, replace=FALSE)
> random_sample_threehundred <- sample(data_age, size=300, replace=FALSE)
> random_sample_fivehundred <- sample(data_age, size=500, replace=FALSE)
> random_sample_onethousand <- sample(data_age, size=1000, replace=FALSE)
> CI_func_tscore_width <- function(sample_data, conf_level) {
+   sample_mean <- mean(sample_data)
+   sample_size <- length(sample_data)
+   sample_sd <- sd(sample_data)
+ 
+   if (sample_size < 2) {
+     return(NA)
+   }
+ 
+   sample_standard_error <- sample_sd/sqrt(sample_size)
+   alpha <- 1 - conf_level
+   critical_t <- qt(1 - (alpha / 2), df = sample_size - 1)
+   sample_moe <- critical_t * sample_standard_error
+ 
+   return(2 * sample_moe)
+ }
> ten_CI_tscore_ninety <- CI_func_tscore_width(random_sample_ten, 0.90)
> ten_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_ten, 0.95)
> ten_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_ten, 0.99)
> fifty_CI_tscore_ninety <- CI_func_tscore_width(random_sample_fifty, 0.90)
> fifty_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_fifty, 0.95)
> fifty_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_fifty, 0.99)
> onehundred_CI_tscore_ninety <- CI_func_tscore_width(random_sample_onehundred, 0.90)
> onehundred_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_onehundred, 0.95)
> onehundred_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_onehundred, 0.99)
> threehundred_CI_tscore_ninety <- CI_func_tscore_width(random_sample_threehundred, 0.90)
> threehundred_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_threehundred, 0.95)
> threehundred_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_threehundred, 0.99)
> fivehundred_CI_tscore_ninety <- CI_func_tscore_width(random_sample_fivehundred, 0.90)
> fivehundred_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_fivehundred, 0.95)
> fivehundred_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_fivehundred, 0.99)
> onethousand_CI_tscore_ninety <- CI_func_tscore_width(random_sample_onethousand, 0.90)
> onethousand_CI_tscore_ninety_five <- CI_func_tscore_width(random_sample_onethousand, 0.95)
> onethousand_CI_tscore_ninety_nine <- CI_func_tscore_width(random_sample_onethousand, 0.99)
> plot_data <- data.frame(
+   sample_size = c(10, 50, 100, 300, 500, 1000,
+                   10, 50, 100, 300, 500, 1000,
+                   10, 50, 100, 300, 500, 1000),
+ 
+   conf_level = as.factor(c(rep(90, 6),
+                            rep(95, 6),
+                            rep(99, 6))),
+ 
+   ci_width = c(ten_CI_tscore_ninety, fifty_CI_tscore_ninety, onehundred_CI_tscore_ninety,
+                threehundred_CI_tscore_ninety, fivehundred_CI_tscore_ninety, onethousand_CI_tscore_ninety,
+ 
+                ten_CI_tscore_ninety_five, fifty_CI_tscore_ninety_five, onehundred_CI_tscore_ninety_five,
+                threehundred_CI_tscore_ninety_five, fivehundred_CI_tscore_ninety_five, onethousand_CI_tscore_ninety_five,
+ 
+                ten_CI_tscore_ninety_nine, fifty_CI_tscore_ninety_nine, onehundred_CI_tscore_ninety_nine,
+                threehundred_CI_tscore_ninety_nine, fivehundred_CI_tscore_ninety_nine, onethousand_CI_tscore_ninety_nine)
+ )
> ggplot(plot_data, aes(x = sample_size, y = ci_width, color = conf_level)) +
+   geom_line(linewidth = 1) +
+   geom_point(size = 3)
>
```



30
20
10
ci_width
0
250
500
750
1000
sample_size
conf_level
90
95
99